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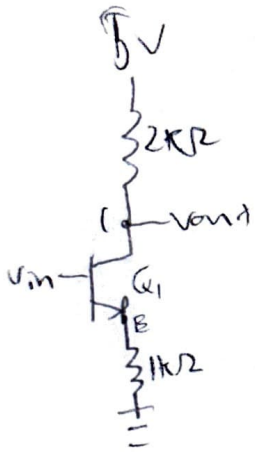
E/EN 252
HW 1

1.



$\beta = 100$

$V_A = 100V$

$100 = \frac{I_C/I_E}{1 - \frac{I_C}{I_E}}$

$100 = 101(I_C/I_E)$

$\frac{100}{101} = \frac{I_C}{I_E}$

$I_C + I_B = I_E$

$I_E(\frac{100}{101}) = I_C$

$I_E(\frac{1}{101}) = I_B$

a. $V_{in} = [0, 0.7]$ cutoff

$V_{in} = (0.7, 2.277]$ active

$V_{in} = (2.277, 5)$ csat

$V_C = V_B + .3$

$V_{in} = V_E + .7$

$V_{CE_{sat}} = .3V$

$\frac{5 - (V_E + .3)}{2k} = \frac{100}{101} I_E$

$\frac{V_E}{1k} = I_E$

$\frac{4.7 - V_E}{2k} = \frac{100 V_E}{101000}$

$\frac{4.7 - V_E}{2} = \frac{100 V_E}{101}$

$474.7 - 101 V_E = 200 V_E$

$V_E = 1.577V$

$1.577 + .7 =$
$2.277V$

b. $V_{in} = 1.7$ so $V_E = 1V$

$\frac{1}{1k} = 1mA = I_E$, $I_C = .9900 mA$, $I_B = .0099 mA$

$\frac{5 - V_{out}}{2k} = .9900 mA$

$V_{out} = 3.0198V$

$g_m = \frac{I_C}{V_T} = \frac{.99 \times 10^{-3}}{\left(\frac{1.38 \times 10^{-23}(300)}{1.602 \times 10^{-19}}\right)} = .0383$ Siemens

$r_\pi = \frac{\beta}{g_m} = 2610.373\Omega$

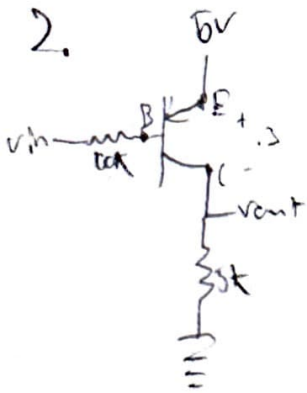
$r_o = \frac{V_A}{I_C} = 101010.101\Omega$

$V_{out} = 3.0198V$, $i_B = .0099 mA \approx .01mA$

$g_m = .0383$ siemens, $2610.373\Omega = r_\pi$,

$r_o = 101010.101\Omega$

2.



I think $V_{BE} = .7V$ is supposed to be $V_{BE} = 1.7V$

$V_B = 4.3V$

$\frac{100}{101} I_E = I_C$

$\frac{1}{101} I_E = I_B$

$5 - .3 = 4.7$

$\frac{4.7}{5k} = 9.4 \times 10^{-4} A = I_C$

$\frac{I_C}{100} = I_B$

$\frac{4.3 - v_{in}}{100k} = 9.4 \times 10^{-6}$

$v_{in} = 3.36$

If $v_{in} > 4.3$, cutoff

a.

$v_{in} = [5, 4.3V)$
cutoff

$v_{in} = (4.3V, 3.36V)$
active

$v_{in} = [3.36V, 0V]$
sat

b.

$\frac{4.3 - 3.8}{100k} = 5\mu A = I_B$ $\quad I_C = 100 I_B = 500\mu A$ $\quad \frac{V_{out}}{5k} = 500\mu A$

$V_{out} = 2.5V$

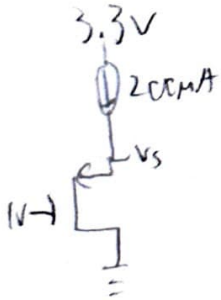
$g_m = \frac{I_C}{V_T} = \frac{5 \times 10^{-4}}{.0258} = .0193 S$

$r_\pi = \frac{\beta}{g_m} = 5168.539 \Omega$

$r_o = \frac{V_A}{I_C} = \frac{100}{5 \times 10^{-4}} = 200000 \Omega$

$V_{out} = 2.5V$
$g_m = .0193 S$
$r_\pi = 5168.539 \Omega$
$r_o = 200k\Omega$

3.



Assume on and test active and sat

active

$200 \times 10^{-6} = 100 \times 10^{-6} \left(\frac{4}{.6}\right)\left((V_6 - V_S + .7)(-V_S) - \frac{(-V_S)^2}{2}\right)$

$.3 = (1.7 - V_S)(-V_S) - \left(\frac{(-V_S)^2}{2}\right)$

$.3 = (V_S^2 - 1.7V_S - \frac{V_S^2}{2})$

$0 = \frac{V_S^2}{2} - 1.7V_S - .3$

$V_S = -.168V$ or 3.568

X not active

sat

$\frac{1}{2}(100 \times 10^{-6})\left(\frac{4}{.6}\right)(V_6 - V_S + .7)^2 = 200 \times 10^{-6}$

$(1.7 - V_S)^2 = .6$

$V_S^2 - 3.4V_S + 2.89 = .6$

$V_S^2 - 3.4V_S + 2.29 = 0$

$V_S = .9254$ or 2.4746

$1 - .9254 + .7 = .7746 > 0$ not this

$1 - 2.4746 + .7 = -.7746 < 0$ ← this

a.

$V_S = 2.4746V$
sat

b. In sat, $g_m = k'\frac{W}{L}V_{GT} = 100\times10^{-6}\left(\frac{4}{16}\right)(-.7746) = -5.164\times10^{-4}\,S$

c. When $V_{GT} = V_{DS}$ that's the barrier

$V_G - V_S + V_T = V_D - V_S$

$1.7 - V_S = -V_S$

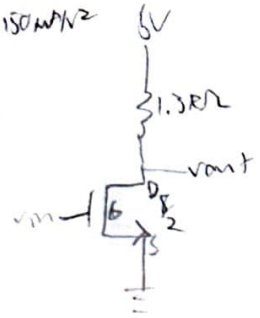
$1.7 = 0$

no

no val possible

4. $W/L = 4$

$k' = 150\,\mu A/V^2$



Vin ≤ .6V cutoff

Vin > .6V and

Vin < 3.121V = sat

Vin > 3.121V = triode

sat

$V_{GT} < V_{DS} \rightarrow V_{GT} < V_{out}$

$I_{Dsat} = \frac{1}{2}k'\frac{W}{L}(V_{GT})^2$

$\frac{5 - V_{out}}{1.3k} = I_D = \frac{1}{2}k'\frac{W}{L}(V_{GT})^2$

$\frac{5 - V_{GT}}{1.3k} = \frac{1}{2}k'\frac{W}{L}(V_{GT})^2$

$5 - V_{GT} = .39\,V_{GT}^2$

$.39V_{GT}^2 + V_{GT} - 5 = 0$

$V_{GT} = 2.521V$ or $-5.085V$

right

$V_{in} = 2.521 + .6 = 3.121$

b. $I_D = \frac{5-3}{1300} = 1.538\,mA$

wish I found this earlier

$V_{ov} = \sqrt{\frac{2I_D}{k'\frac{W}{L}}} = 2.265V$

c. $g_m = k'\frac{W}{L}V_{ov} = 1.359\,mS$

$r_o = \frac{1}{\lambda I_D} = \frac{1}{1.538\times10^{-3}(.02)} = 32500\,\Omega$

d. Small sig gain $(A_v?) = -g_m(R_D \| r_o) = -1.698\,V/V$

Output impedance $(Z_{out}) = 32500 \| 1300 = 1250\,\Omega$